

Operating and service manual

Addendum A2

STAR  COOL

**Addendum for
Operating and service manual
Version 810900A**

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1 Introduction

This user's manual addendum is valid for software version 0235 or newer versions.

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2 Sections with changes

2.1 Section 4: General description

Old text:

The unit is equipped with a de – humidification function controlled by the electronic controller of the unit. The humidity setpoint can be set in the range from 95 – 60 % RH. The unit can control to the lowest level. The de – humidification function is active as long as the temperature control is in set point range. The unit is equipped with heating elements, mounted under the evaporator coil, for the dehumidification. The de – humidification system is disabled in economy mode.

New text:

The unit is equipped with a de – humidification function controlled by the electronic controller of the unit. The humidity setpoint can be set in the range from 95 – 60 % RH. The unit can control to the lowest level. The de – humidification function is active as long as the temperature control is in set point range. The unit is equipped with heating elements, mounted under the evaporator coil, for the de – humidification. The de – humidification system is also active in economy mode.

2.2 Section 5.2.8: Evaporator Fan

Old text:

The evaporator fan function ensures correct fan speed.
The function is active in the automatic mode.

This function has 2 modes:

- Normal
- Economy

Normal

In the chill mode the fans run at high speed.
In the frozen mode the fans run at low speed.

Economy

The fans run at a constant low speed. By selecting Economy mode, dehumidification is turned off.

New text:

The evaporator fan function ensures correct fan speed.
The function is active in the automatic mode.

This function has 2 modes:

- Normal
- Economy

Normal

In the chill mode the fans run at high speed.
In the frozen mode the fans run at low speed.

Economy

The fans runs at constant low speed to save energy.
The evaporator fan function switches to use economy mode whenever:

- a) The setpoint is > 0°C (32F)
- b) Fresh air exchange valve is closed
- c) Dehumidification is turned off.

Or

The operator switches manually to economy mode.

Normal mode is disabled when the above mentioned conditions a), b) and c) are all fulfilled.

2.3 Section 5.3.1: Function Test

Old text:

8. Compressor/valve test (Vexpansion, Vhotgas and Veconomizer)

New text:

8. Compressor/FC/valve test (Frequency Converter, Vexpansion, Vhotgas and Veconomizer)

2.4 Section 11.3: Temperature Setting

Old text:

Upon reaching desired temperature, press . The set-point will be accepted and shown in normal writing again.

New text:

Upon reaching desired temperature, press  and hold for 3 seconds. The new set-point will be accepted and displayed.

2.5 Section 11.6: PTI or Function Test Execution

T11 Test result: 80 Compressor:

Old text:

**T11 Test result: 80 Compressor
(80 Compressor)**

Function:

Show the status of the test of the compressor.

Value:

“_” if not done yet. “RUN” if still running test. “PASS” if test finished successfully. “FAIL” if the test failed. “ABORT” if the test was aborted by the user. See specific description for alarm 845 in section 17.8.

New text:

**T11 Test result: 80 Compressor
(80 Compressor)**

Function:

Show the status of the test of the compressor.

Value:

“_” if not done yet.

“RUN” if still running test.

“PASS” if test finished successfully.

“FAIL” if the test failed.

“ABORT” if the test was aborted by the user. See specific description for alarm 845 and 846 in section 17.8.

2.6 Section 11.8: Operation Parameter Setting

Old Text:

**O03 Airflow mode
(Airflow mode)**

Function:

Setting the control modes: Normal or economy mode.

If economy mode is selected:

- Evaporator fan will operate at low speed in the chill mode.
- Dehumidification is turned off!

If set point temperature is $> 0^{\circ}\text{C}$ (32°F), and air exchange is closed and humidity setpoint = off, the evaporator fan is forced to Economy mode. Economy mode can be overridden with selecting Normal mode in this menu.

Value:

Function set to NORMAL, ECONOMY. Default is NORMAL.

New text:**O03 Airflow mode
(Airflow mode)****Function:**

Setting the control modes: NORMAL or ECONOMY mode.

If NORMAL mode is selected:

- In the chill mode the fans run at high speed.
- In the frozen mode the fans run at low speed.

If ECONOMY mode is selected:

- Evaporator fans runs at constant low speed.

The controller also switches to use economy mode whenever:

- The setpoint is $> 0^{\circ}\text{C}$ (32°F)
- Fresh air exchange valve is closed
- Dehumidification is turned off

Or

The operator manually switches to ECONOMY mode.

NORMAL mode is disabled when the above mentioned conditions a), b) and c) are all fulfilled.

Value:

Function set to NORMAL, ECONOMY. Default is NORMAL.

2.7 Section 11.9: Alarms

Alarm 846 has been added to the alarm list:

8. Test alarms			
846	FC check	FC internal fault	Alarm

2.8 Section 11.11: $^{\circ}\text{C}$ and $^{\circ}\text{F}$ Temperature Scale

Old text:

If the display shows $^{\circ}\text{C}$:

By pressing  shortly $^{\circ}\text{F}$ is shown for 10 sec.; then display switches to $^{\circ}\text{C}$ showing.

By pressing and holding  for 3 sec. $^{\circ}\text{F}$ is constantly shown.

If the display shows $^{\circ}\text{F}$:

By pressing  shortly $^{\circ}\text{C}$ is shown for 10 sec.; then display switches to $^{\circ}\text{F}$ showing.

By pressing and holding  for 3 sec. $^{\circ}\text{C}$ is constantly shown.

Note that temperature and pressure showing go together. When $^{\circ}\text{C}$ is shown, pressure is displayed in units of Bar, and when $^{\circ}\text{F}$ is shown pressure is displayed in units of PSI.

Shown pressure is relative to atmosphere pressure.

New text:

As long as the button  is pressed, $^{\circ}\text{F}$ is shown instead of $^{\circ}\text{C}$ and PSI is shown instead of Bar. When the button is released, $^{\circ}\text{C}$ and Bar is shown again.

Permanently change to $^{\circ}\text{F}/\text{PSI}$ is not available.

Shown pressure is relative to atmosphere pressure.

2.9 Section 17.1: Temperature Sensor Alarms

Alarm 105 Tsup1 invalid:

Old text:

105	Tsup 1 invalid					Warning
Description	Supply Air Temperature Sensor 1 Invalid.					
Criteria	Value below alarm limit -58F (-50°C) or above +158F (+70°C). Value invalid for 30 sec. for alarm activation.					
Cause	Indication of defective supply air temperature sensor or its measuring circuitry.					
Action	Replacement by new value from AAS system.					
	Log	X	Alarm	X	Alarm light	Off
Consequence	Deteriorated control precision in the chill mode.					
Elimination	When sensor value becomes valid, it is marked as inactive in the alarm list and may then be deleted. Value must be valid for 60 sec. to set alarm inactive.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive					
Trouble shooting	<u>Possible causes:</u>					
	- Active alarms AL103 or AL104. - Temperature sensor reading is out of valid range: -58F (-50°C) or above +158F (+70°C).					
	<u>Accompanied alarms:</u>					
	AL 103 or 104 may also be active.					
	<u>Trouble shooting:</u>					
	1) If alarms AL103 or AL104 are active, check their trouble shooting section first. 2) Disconnect the sensor cable for sensor Tret from the connector on the interface PCB, according to the wiring schematics inside in the control cabinet. 3) Measure the resistance between the two wires. a) If the resistance is out of range of the resistance and temperature table in section 26.1, the temperature sensor and cable are defect and should be replaced.					

New text:

105	Tsup 1 invalid					Warning
Description	Supply Air Temperature Sensor 1 Invalid.					
Criteria	Value is below alarm limit -50°C (-58F) or above +70°C (+158F) or difference between Tsup1 and Tsup2 is > 1°C for 30 min. Value invalid for 30 sec. for alarm activation.					
Cause	Indication of defective supply air temperature sensor or its measuring circuitry or sensor not mounted correctly in unit.					
Action	Replacement by new value from AAS system.					
	Log	X	Alarm	X	Alarm light	Off
Consequence	Deteriorated control precision in the chill mode.					
Elimination	When sensor value becomes valid, it is marked as inactive in the alarm list and may then be deleted. Value must be valid for 60 sec. to set alarm inactive.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive					
Trouble shouting	<u>Possible causes:</u>					
	- Active alarms AL103 or AL104. - Temperature sensor reading is out of valid range: -50°C (-58F) or above +70°C (+158F). - Difference between Tsup1 and Tsup2 is larger than 1 °C: 1 °C difference for more than 30 min up to 10 °C difference in more than 3 min.					
	<u>Accompanied alarms:</u>					
	AL 103 or 104 may also be active.					
	<u>Trouble shooting:</u>					
	- If alarms AL103 or AL104 are active, check their trouble shooting section first. - Check that both sensors, Tsup1 and Tsup2 are mounted correct in their mounting holes. - Disconnect the sensor cable for sensor Tsup1 from the connector on the interface PCB, according to the wiring schematics inside in the control cabinet. - Measure the resistance between the two wires. If the resistance is out of range of the resistance and temperature table in section 26.1, the temperature sensor and cable are defect and should be replaced.					

Alarm 108 Tsup2 invalid:**Old text:**

108	Tsup 2 invalid					Warning
Description	Supply Air Temperature Sensor 2 Invalid.					
Criteria	Value below alarm limit -58F (-50°C) or above +158F (+70°C). Value invalid for 30 sec. for alarm activation.					
Cause	Indication of defective supply air temperature sensor or its measuring circuitry.					
Action	Replacement by new value from AAS system.					
	Log	X	Alarm	X	Alarm light	Off
Consequence	Deteriorated control precision in the chill mode.					
Elimination	When sensor value becomes valid, it is marked as inactive in the alarm list and may then be deleted. Value must be valid for 60 sec. to set alarm inactive.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive					
Trouble shouting	<u>Possible causes:</u>					
	- Active alarms AL106 or AL107. - Temperature sensor reading is out of valid range: -58F (-50°C) or above +158F (+70°C).					
	<u>Accompanied alarms:</u>					
	AL 106 or 107 may also be active.					
	<u>Trouble shooting:</u>					
	1) If alarms AL106 or AL107 are active, check their trouble shooting section first. 2) Disconnect the sensor cable for sensor Tret from the connector on the interface PCB, according to the wiring schematics inside in the control cabinet. 3) Measure the resistance between the two wires. a) If the resistance is out of range of the resistance and temperature table in section 26.1, the temperature sensor and cable are defect and should be replaced.					

New text:

108	Tsup 2 invalid					Warning
Description	Supply Air Temperature Sensor 2 Invalid.					
Criteria	Value is below alarm limit -50°C (-58F) or above +70°C (+158F) or difference between Tsup1 and Tsup2 is > 1°C for 30 min. Value invalid for 30 sec. for alarm activation.					
Cause	Indication of defective supply air temperature sensor or its measuring circuitry or sensor not mounted correctly in unit.					
Action	Replacement by new value from AAS system.					
	Log	X	Alarm	X	Alarm light	Off
Consequence	Deteriorated control precision in the chill mode.					
Elimination	When sensor value becomes valid, it is marked as inactive in the alarm list and may then be deleted. Value must be valid for 60 sec. to set alarm inactive.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive					
Trouble shouting	<u>Possible causes:</u>					
	- Active alarms AL106 or AL107. - Temperature sensor reading is out of valid range: -50°C (-58F) or above +70°C (+158F). - Difference between Tsup1 and Tsup2 is larger than 1 °C: 1 °C difference for more than 30 min up to 10 °C difference in more than 3 min.					
	<u>Accompanied alarms:</u>					
	AL 106 or 107 may also be active.					
	<u>Trouble shooting:</u> - If alarms AL106 or AL107 are active, check their trouble shooting section first. - Check that both sensors, Tsup1 and Tsup2 are mounted correct in their mounting holes. - Disconnect the sensor cable for sensor Tsup2 from the connector on the interface PCB, according to the wiring schematics inside in the control cabinet. - Measure the resistance between the two wires. If the resistance is out of range of the resistance and temperature table in section 26.1, the temperature sensor and cable are defect and should be replaced.					

2.10 Section 17.5: Frequency Converter Alarms

Alarm 516, FC TRIP phase loss:

Old text:

516	FC TRIP phase loss					Alarm
Description	Frequency Converter Phase Loss Fault Alarm.					
Criteria	Frequency converter cannot maintain DC filter voltage.					
Cause	Indication of defective frequency converter or lack of phase.					
Action	Frequency converter shut-down.					
	Log	X	Alarm	X	Alarm light	Slow flash
Consequence	Unit stops.					
Elimination	Alarm will be marked as inactive in alarm list when reset by frequency converter. Alarm may then be deleted.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive	Freq FC	I FC	Psuc	Pdis	
Trouble shooting	<u>Possible causes:</u>					
	- One or more phases are not applied to the FC. - Defect FC.					
	<u>Accompanied alarms:</u>					
	AL 417, 418, 419, 523 may be active.					
	<u>Trouble shooting:</u>					
	1) Verify that there is correct voltages for all 3 phases to the FC.					
	2) Supply unit with correct power voltage according to specification.					
	If there is no FC replacement available, the unit can be rewired for emergency operation.					

New text:

516	FC TRIP phase loss					Alarm
Description	Frequency Converter Phase Loss Fault Alarm.					
Criteria	Frequency converter cannot maintain DC filter voltage (or too much ripple in DC voltage).					
Cause	Indication of lack of phase or defective frequency converter.					
Action	Frequency converter shut-down.					
	Log	X	Alarm	X	Alarm light	Slow flash
Consequence	Unit stops.					
Elimination	Alarm will be marked as inactive in alarm list when reset by frequency converter. Alarm may then be deleted.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive	Freq FC	I FC	Psuc	Pdis	
Trouble shooting	<u>Possible causes:</u>					
	- One or more phases are not applied to the FC. - Unstable power supply (generator). - Defect FC.					
	<u>Accompanied alarms:</u>					
	AL 417, 418, 419, 523 may be active.					
	<u>Trouble shooting:</u>					
	1) Verify that there is correct voltages for all 3 phases to the FC					
	2) Verify that voltage levels are the same for all 3 phases.					
	3) Supply unit with correct power voltage according to specification.					
	4) Replace the FC.					
	If there is no FC replacement available, the unit can be rewired for emergency operation.					

Alarm 523, FC phase loss:**Old text:**

523	FC phase loss					Warning
Description	Frequency Converter Phase Loss Fault Warning.					
Criteria	Frequency converter cannot maintain DC filter voltage above warning limit 440 VDC.					
Cause	Indication of defective frequency converter or lack of phase.					
Action	None.					
	Log	X	Alarm		Alarm light	Off
Consequence	If not solved, this problem can cause frequency converter to stop due to alarm 516.					
Elimination	Warning will be marked as inactive in alarm list when reset by frequency converter. Alarm may then be deleted.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive	Freq FC	I FC	Psuc	Pdis	
Trouble shooting	<u>Possible causes:</u>					
	- One or more phases are not applied to the FC. - Defect FC.					
	<u>Accompanied alarms:</u>					
	AL 417, 418, 419, 516 may be active.					
	<u>Trouble shooting:</u>					

- 1) Verify that there is correct voltages for all 3 phases to the FC.
- 2) Supply unit with correct power voltage according to specification.

If there is no FC replacement available, the unit can be rewired for emergency operation.

New text:

523	FC phase loss					Warning
Description	Frequency Converter Phase Loss Fault Warning.					
Criteria	Frequency converter cannot maintain DC filter voltage above warning limit 440 VDC (or too much ripple in DC voltage).					
Cause	Indication of defective frequency converter or lack of phase.					
Action	None.					
	Log	X	Alarm		Alarm light	Off
Consequence	If not solved, this problem can cause frequency converter to stop due to alarm 516.					
Elimination	Warning will be marked as inactive in alarm list when reset by frequency converter. Alarm may then be deleted.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive	Freq FC	I FC	Psuc	Pdis	
Trouble shooting	<u>Possible causes:</u>					
	- One or more phases are not applied to the FC. - Unstable power supply (generator). - Defect FC.					
	<u>Accompanied alarms:</u>					
	AL 417, 418, 419, 516 may be active.					
	<u>Trouble shooting:</u>					

- 1) Verify that there is correct voltages for all 3 phases to the FC.
- 2) Verify that voltage levels are the same for all 3 phases.
- 3) Supply unit with correct power voltage according to specification.
- 4) Replace the FC.

If there is no FC replacement available, the unit can be rewired for emergency operation.

2.11 Section 17.6: Operation Alarms

Alarm 613, Motor Trouble:

Old text:

613	Motor trouble					Alarm	
Description	Evaporator Motor 1 or 2 Overheated several times or permanently within short time.						
Criteria	There have been several overheating or a permanent overheating signal from evaporator motor 1 or 2 within 30 min.						
Cause	One or both evaporater motors are overheating either permanently or repeatedly generated overheat alarm within the last 30 min.						
Action	Restart delay increased to 10-12 min.						
	Log	X	Alarm	X	Alarm light	Slow flash	
Consequence	There is no or limited air circulation in the container.						
Elimination	When the evaporator motors have been without errors for more than 1 hours, this alarm is marked inactive and can be deleted.						
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5		
	Active/Inactive						
Trouble shooting	<u>Possible causes:</u>						
	Repeated evaporator motor trouble.						
	<u>Accompanied alarms:</u>						
	Alarms AL 400 or AL 401 are/have been active.						
	<u>Trouble shooting:</u>						
	1) Check and clear AL 400 or AL 401 first.						
	2) Start a manual defrost to remove the ice there may be in the motor(s).						

New text:

613	Motor trouble				Alarm	
Description	Evaporator fan Motor 1 or 2 or condenser fan motor overheated several times.					
Criteria	There have been several overheating or a permanent overheating signal from evaporator fan motor 1 or 2 or condenser fan motor within 30 min.					
Cause	One or both evaporator fan motors or the condenser fan motor are overheating either permanently or repeatedly generated overheat alarm within the last 30 min.					
Action	Restart delay increased to 10-12 min.					
	Log	X	Alarm	X	Alarm light	Slow flash
Consequence	Evaporator fan motor: There is no or limited air circulation in the container. Condenser fan motor: There is limited capacity in the condenser.					
Elimination	When the evaporator fan motors and the condenser fan motor have been without errors for more than 1 hour, this alarm is marked inactive and can be deleted.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive					
Trouble shooting	<u>Possible causes:</u> - Repeated evaporator fan motor trouble. - Repeated condenser fan motor trouble.					
	<u>Accompanied alarms:</u> Alarms AL 400, AL 401 or AL 402 are/have been active.					
	<u>Trouble shooting:</u>					
	1) Check and clear AL 400, AL401 or AL 402 first.					
	2) Check that the condenser fan can rotate freely.					
	3) Check that the evaporator fans can rotate freely.					
	4) Change the defect motor.					

2.12 Section 17.8: Test Alarms

New alarm 846 FC Check:

New text:

846	FC Check					Alarm
Description	Internal fault in FC.					
Criteria	FC temperature not increased with 15°C within 5 min during step 8 of Function test					
Cause	Indication of internal failure in the FC.					
Action	None					
	Log	X	Alarm	X	Alarm light	Slow flash
Consequence	Test failed.					
Elimination	Alarm may then be deleted after test completed.					
Log data	Parm 1	Parm 2	Parm 3	Parm 4	Parm 5	
	Active/Inactive	T0	Psuc	Peco	Pdis	
Trouble shooting	<u>Possible causes:</u> Internal failure in the FC					
	<u>Accompanied alarms:</u> N/A.					
	<u>Trouble shooting:</u>					
	1) Check and clear other alarms first.					
	2) Check that FC cover is mounted correctly with all screws.					
	3) Check motor cable (Connection cable between FC and compressor).					
4) The frequency converter needs repair and should be replaced.						

2.13 Section 26.5: Airexchange Sensor table

New table entries for Air exchange.

Vout [Volt]	Air exchange [m ³ /h]
0.0	0
0.1	10
0.2	20
0.3	30
0.4	40
0.5	50
0.6	60
0.7	70
0.8	75
0.9	85
1.0	95
1.1	100
1.2	110
1.3	115
1.4	120
1.5	130
1.6	135
1.7	140
1.8	145
1.9	145
2.0	150
2.1	155
2.2	160
2.3	160
2.4	165
2.5	165
2.6	170
2.7	170
2.8	175
2.9	180
3.0	180
3.1	185
3.2	190
3.3	195
3.4	200
3.5	205
3.6	210
3.7	220